

SAME EVIDENCE, DIFFERENT SKILL: DIAGNOSING STATE-GENERATION VARIANCE IN SELF-EVOLVING AGENTS

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Paper under double-blind review

ABSTRACT

Self-evolving agents turn execution trajectories into persistent skills, prompts, or memories that alter future behavior. Existing evaluations usually vary which feedback the updater sees while treating the update mechanism itself as a stable conduit from evidence to state. We show that this assumption can fail. In a controlled SpreadsheetBench skill-evolution substrate, a 48-pair study that replaces winner evidence with a deterministic failed nonwinner yields a small positive mean effect (+2.31 points) but neither superiority nor practical equivalence, with substantial stream heterogeneity. We then localize the instability. A historically strong learned state remains useful when frozen and re-evaluated, but replaying the byte-identical updater evidence does not regenerate the historical effect: two fresh updater realizations yield 0 and -5.56 points relative to matched winner-only states. Thus, in this development case, the evidence is not the state: persistent-state synthesis is itself a candidate source of variance. We formalize the resulting causal chain as evidence selection $\rightarrow$ state generation $\rightarrow$ persistent-state semantics $\rightarrow$ downstream behavior, and introduce a variance-controlled typed state compiler that maps learner-visible trajectory evidence to a finite repair representation before canonical compilation. A prospective 2×2 bridge crosses evidence source (Winner vs. First-Fail-4) with state generator (native free-form updater vs. constrained compiler), with a score-matched trajectory-blind falsifier and disjoint SCREEN/VALIDATION panels. The paper’s central claim is therefore not that failures are inherently better feedback, nor that structured diagnosis is novel, but that state-generation realization is an independently testable bottleneck in persistent agent self-evolution.

1 INTRODUCTION

Self-evolving agents persist experience as skills, prompts, or memories that alter future behavior. A common abstraction is evidence E entering an updater U , which produces persistent state S for a future policy π : $E \rightarrow U \rightarrow S \rightarrow \pi \rightarrow Y$. Existing evaluations mostly intervene on E —success versus failure feedback, different history windows, or different retrieved trajectories—while treating U as a stable conduit. But when S is itself free-form generated text, that assumption need not hold: the same evidence can be rewritten with different scope, redundancy, or procedural specificity, and those state realizations may induce different behavior.

We encounter this issue while studying best-of- K skill evolution. Search generates several trajectories; a verifier selects one winner for acting; a persistent updater later receives a projection of the same frozen pool. Our initial question was whether the trajectory worth serving is necessarily the trajectory worth learning from. Holding the search pool, served winner, initial skill, updater, and evaluation harness fixed, we replaced winner evidence with a deterministic rejected witness. Across 48 matched pairs, the mean effect was only +2.31 percentage points, with substantial stream heterogeneity; the preregistered superiority test did not pass. Thus the simple claim that rejected failures are universally better learning evidence is unsupported.

We then used the negative result to localize the bottleneck rather than tuning the evidence selector until it passed. A progress-based “diagnostic” witness selector failed its frozen continuation gate. In contrast, a historically strong First-Fail state remained directionally useful when frozen and re-

evaluated. Most importantly, reconstructing the *byte-identical updater evidence* did not regenerate the historical advantage: two fresh updater realizations produced matched downstream contrasts of 0/18 and $-1/18$. The evidence was the same; the generated persistent state was not equivalently useful.

This observation motivates the causal decomposition

$$T_K \xrightarrow{g} E \xrightarrow{U} S \xrightarrow{\pi} Y, \quad (1)$$

where T_K is the generated pool, g is the learning projection, and U is an explicit experimental factor rather than hidden plumbing. Evidence selection and state synthesis are different causal objects. Conflating them can make a lucky state realization look like an evidence effect, or make useful evidence look ineffective because the updater encoded it poorly.

We therefore study *variance-controlled persistent-state generation*. At the state level, a deterministic repair ladder tests whether compact procedural semantics are sufficient. At the method level, a prospective 2×2 bridge crosses evidence source (Winner vs. First-Fail-4) with state generator (native free-form updater vs. a typed canonical compiler). A score-matched trajectory-blind control asks whether the compiler merely notices failures and injects generic workflow hygiene. SCREEN and VALIDATION use disjoint heldout panels, and failure of an upstream gate cannot be rescued by opening E3 or another backbone.

The novelty is deliberately narrow. Rethinking Self-Evolving Agent Skills already establishes feedback-dependent and heterogeneous skill evolution (Liu et al., 2026b); SkillRevise already performs execution-grounded diagnosis and repair-principle-guided skill revision (Liu et al., 2026a); WikiSkill already separates raw experience, persistent knowledge, and executable skills (Tang et al., 2026). We do not claim “failures help,” structured diagnosis, or experience-to-knowledge compilation as new. Our residual object is whether *state-generation realization itself* is an independently identifiable bottleneck under matched evidence, and whether controlling that interface changes future utility.

Contributions. (1) We separate learning projection, state generation, persistent-state semantics, and downstream execution in a causal model of skill self-evolution. (2) We provide a mechanism-localization chain in which a strong frozen state survives remeasurement but byte-identical evidence fails to regenerate its historical effect. (3) We introduce a canonical typed state compiler and a prospective Evidence $\times$ Generator experiment that tests generator and evidence effects separately. (4) We preregister falsifiers and decision rules so that a compiler win without a failure-evidence win remains a state-compilation result rather than being promoted into a rejected-trajectory claim.

2 PROBLEM SETTING AND CAUSAL DECOMPOSITION

For task x and initial persistent state S_0 , search produces a frozen pool $T_K = \{\tau_1, \dots, \tau_K\}$ with deterministic verifier scores $r_i \in \{0, 1\}$. Acting uses a fixed projection $a(T_K)$: the highest-scoring trajectory, with original rollout order as tie-breaker. Learning independently receives a projection $g(T_K)$, rendered as an exact learner-visible package

$$E = \mathcal{R}(g(T_K)) = (\text{trajectory text, selected verifier score}). \quad (2)$$

The updater produces persistent state $S = U(E, S_0; \xi_U)$, and unseen task q is evaluated as $Y(q; S) = \text{Verify}(\pi(q, S; \xi_A))$. Here ξ_U and ξ_A denote residual updater and actor realization under the frozen provider configuration.

This factorization yields three distinct questions. First, holding U fixed, an *evidence-package effect* compares $U(E_1)$ with $U(E_0)$. Second, holding E fixed, a *state-generation-method effect* compares two complete generators, e.g. $U_{\text{comp}}(E)$ and $U_{\text{free}}(E)$. Third, repeated synthesis from identical E exposes *state-realization variation*. Because the verifier score is part of the updater-visible record, our evidence intervention concerns a trajectory+score package; it does not identify trajectory text separately from score.

This distinction matters whenever persistent state is generated language. If $Y(U(E_{\text{fail}})) > Y(U(E_{\text{win}}))$, either the failure package carried more reusable information or its sampled state realization happened to be better. Conversely, useful evidence can appear ineffective when a noisy

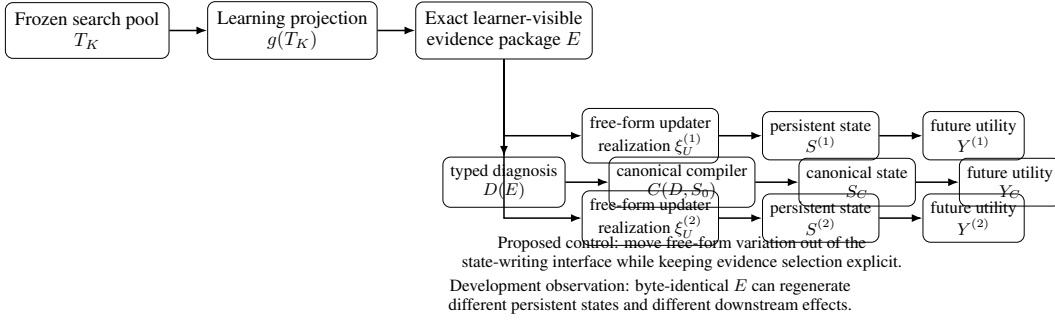


Figure 1: **The evidence is not the state.** Our causal decomposition exposes persistent-state generation as a separate experimental factor. The top branch illustrates the exact-evidence replay: the same learner-visible evidence can enter fresh free-form updater realizations and yield different downstream behavior. The bottom branch shows the proposed variance-controlled interface, where evidence is mapped to a typed diagnosis and compiled canonically. The paper does not assume that the compiler is better; the prospective bridge tests that claim.

Stage	Intervention	Result	Localization
Repair2	MRW vs. WIN-C, 48 pairs	mean +2.31 pp; $p = .171875$; 95% boot. CI $[-1.85, +6.60]$	no stable global evidence-source superiority
S1	WIN / First-Fail / Progress-Fail / Contrast	13/18, 17/18, 14/18, 14/18	proposed diagnostic selector fails
Frozen state	remeasure historical strong state	FF vs. WIN: 15–14, 16–12	useful state persists without updater
Exact replay	byte-identical evidence, fresh updater	FF vs. WIN: 15–15, 11–12	evidence does not regenerate historical utility

Table 1: Mechanism-localization chain. The last three stages are development-only and localize the next causal intervention; they are not pooled into a confirmatory treatment estimate.

state writer encodes it poorly. The generated state must therefore be measured rather than treated as invisible implementation detail.

We maintain three evidence levels. *State-level mechanism* experiments intervene on frozen states; *automatic method* experiments prospectively convert evidence into state; *confirmatory generalization* uses an untouched panel after method selection. The current G0–G3 study belongs to the first level, the Evidence×Generator bridge to the second, and untouched E3 remains sealed for the third.

3 LOCALIZING THE BOTTLENECK

The mechanism was not inferred from one positive run. A sequence of frozen development experiments progressively removed simpler explanations (Table 1).

Evidence selection alone is unresolved. The closed study contains 48 matched update pairs, 96 learned states, and 1728 heldout units. WIN-C exposes the served winner; MRW uses the deterministic failed-nonwinner projection under the frozen mixed-pool rule while preserving the same acting winner. Mean MRW–WIN-C is 0.02315 (SD 0.07797); the preregistered one-sided sign-flip test gives $p = .171875$ and the bootstrap interval crosses zero. The immutable verdict is `HOLD_MRW_UNDERPOWERED_OR_HETEROGENEOUS`. This rejects a universal rejected-witness story while preserving heterogeneity as a fact to explain.

A richer witness selector does not rescue the story. On one explicitly development-only stream, Progress-Fail and Progress-Contrast each reached 14/18, below the historical First-Fail state at 17/18; the frozen continuation gate therefore stopped. We retain this failed intervention rather than tuning a witness score after outcomes.

The strong state survives when the updater is removed. We froze the historical First-Fail state and a matched WIN-C state and re-evaluated without state generation. Across two fresh replicates, First-Fail achieved 15/18 versus 14/18 and 16/18 versus 12/18. This is selected-case evidence, but it shows that the useful behavior can persist once the state itself is held fixed.

Byte-identical evidence does not regenerate the strong state. We reconstructed the exact learner-visible evidence that produced the strong development state; all 16 rendered packet hashes matched the originals. Two fresh paired updater realizations then yielded 15/18 versus 15/18 and 11/18 versus 12/18, rather than the historical +4/18 contrast. We therefore record a frozen *state-generation replication failure*. With only two fresh realizations this is not a population variance estimate. It is a localization result: evidence bytes are insufficient to characterize the treatment consumed by the future actor, so state generation becomes an explicit experimental variable. The full machine-readable verdict identifier is preserved in Appendix A.

The supportable claim at this stage is deliberately local: on a selected development case, a useful persistent state remained useful when frozen, whereas byte-identical evidence failed to reproduce the historical effect through fresh free-form synthesis. This motivates, but does not prove, a state-generation bottleneck.

4 VARIANCE-CONTROLLED STATE GENERATION

We intervene on the state-writing interface while preserving the learner-visible information set. Both generator arms receive the same eight rendered packages $E_i = (\text{trajectory text}_i, r_i)$. The constrained path cannot use arm, family, task/template identity, hidden labels, golden answers, heldout outcomes, or historical development results. A pre-execution parity audit binds its evidence hashes to those seen by the corresponding free-form updater.

Typed diagnosis. A deterministic parser extracts only visible execution signals: whether the trajectory reads the workbook, materializes a write, saves output, reloads and verifies output, encounters a tool error, and visibly recovers. These signals map to a small frozen diagnosis vocabulary with three repair primitives:

$$\text{VERIFY_OUTPUT}, \quad \text{COMPLETE_WORKFLOW}, \quad \text{RECOVER_TOOL_ERROR}. \quad (3)$$

Successful selected trajectories do not automatically receive generic advice. For failures, missing materialization/save requests completion; saved-but-unverified output requests verification; and an unrecovered visible tool error requests recovery. The vocabulary is a frozen development object, not a universal error ontology.

Canonical compilation. Given diagnoses $\{d_i\}_{i=1}^8$, the compiler produces

$$S_C = C(S_0, \{d_i\}_{i=1}^8) \quad (4)$$

by taking the union of requested primitives, rendering them in fixed order, deduplicating repeated evidence, and removing redundant surfaces (the completion loop already includes reload/verification). Identical diagnosis bundles compile byte-identically. Zero-provider tests reproduce the exact frozen G1/G2/G3 state surfaces for corresponding diagnoses, connecting the automatic method to the earlier state-level intervention.

The comparison with the native updater is a *complete-method* effect, not an isolated effect of determinism: free-form synthesis can also differ in wording, length, scope, and redundancy. We therefore report state bytes/tokens and clause counts but do not post-hoc truncate free-form states to manufacture length matching.

Strong generic falsifier. A compiler may appear evidence-conditioned while merely noticing failure scores and injecting good spreadsheet hygiene. We freeze `SCORE_ONLY_GENERIC_MAX`, which receives the same ordered eight binary selected scores but no trajectory text. All-success input returns S_0 ; any failure compiles the strongest nonredundant generic v1 bundle. A trajectory-conditioned diagnosis claim therefore requires `FF4_COMP` to exceed this score-matched control. If not, at most a generic canonical-state result remains.

Implementation details, exact primitive texts, state hashes, and static compiler tests are provided in Appendix A.

5 EXPERIMENTS

Our controlled substrate uses the first-party MindMemOS SkillEvolver and deterministic SpreadsheetBench-style tasks (Liang et al., 2026). Search pools, evidence renders, learned states, verifier inputs, and provider budgets are content-addressed; completion audits precede score access. Exact runtime/model bindings are deferred to Appendix A.

5.1 M2: STATE-SEMANTIC SUFFICIENCY

Before evaluating an automatic compiler, we ask whether the candidate persistent-state semantics are sufficient locally. On the selected development stream we freeze: G0 (initial skill), G1 (output reload/verification), G2 (Inspect→Read→Compute→Write→Save→Reload/Verify), and G3 (G2 plus clean tool-error recovery). Each is evaluated on the same 18 development heldout tasks. The primary gate selects the simplest $G1 < G2 < G3$ arm exceeding G0 by at least 1/18; a pre-outcome 17-task provider-era sensitivity can veto but never rescue a PASS.

At this revision, 45/72 state-task units are validly complete and the remaining 27 are frozen for exactly-once recovery after a hosted quota reset. No partial scores have been opened. If M2 passes, only G0 and the simplest passing arm receive exactly two complete frozen-state remeasurements; failure stops the branch before automatic method evaluation.

5.2 M4: PROSPECTIVE EVIDENCE × GENERATOR BRIDGE

Only after M2 and its stability gate survive do we authorize the automatic bridge:

	Native free-form updater	Constrained compiler
Winner	W_FREE	W_COMP
First-Fail-4	FF4_FREE	FF4_COMP

All cells start from the same S_0 and frozen search pools; learning cannot alter the served acting winner. Winner exposes the served winner package on all eight update pools. First-Fail-4 deterministically replaces exactly four eligible mixed-pool winner packages with the lowest-index failed nonwinner and leaves the other four unchanged. This equalizes replacement count, not information content.

Fresh evaluation. The bridge uses a fresh b7–b9 suite with zero task-ID and XLSX-hash overlap with the earlier controlled suite and no use of untouched E3 b5/b6. Its 120 formal tasks comprise 96 update tasks (12 streams×8), 12 SCREEN heldout tasks, and 12 disjoint VALIDATION heldout tasks. Six family-balanced streams are frozen for each stage. Suite construction and collision audits are Appendix A.

For stream utility J , the primary contrasts are

$$G_F = J(\text{FF4_COMP}) - J(\text{FF4_FREE}), \quad (5)$$

$$S_C = J(\text{FF4_COMP}) - J(\text{W_COMP}), \quad (6)$$

$$I = G_F - [J(\text{W_COMP}) - J(\text{W_FREE})]. \quad (7)$$

SCREEN passes only if mean $G_F > 0$ and $S_C > 0$, each is positive on at least 4/6 streams, mean $I > 0$, FF4_COMP exceeds SCORE_ONLY_GENERIC_MAX, and no parity/technical violation occurs. Otherwise VALIDATION remains sealed. If SCREEN passes, the unchanged method runs on the disjoint VALIDATION streams/panel under the same directional conditions. No threshold, primitive, compiler rule, or replacement dose can be changed between stages.

Decision logic. A failed generator contrast stops the state-generation method story. A positive generator contrast with failed source contrast supports at most reliable state compilation, not rejected-failure learning. A failed score-only falsifier removes trajectory-conditioned diagnosis. Only survival of generator, source, falsifier, and interaction gates makes a separate untouched-E3 proposal eligible; it does not automatically authorize E3.

Stage	Object	Status	Scientific role
M0	48-pair evidence study	Complete: HOLD	evidence-source heterogeneity
M1	localization chain	Complete; development	candidate generator bottleneck
M2	G0/G1/G2/G3	45/72; scores sealed	state-semantic sufficiency
M3	frozen-state stability	not authorized	local stability
M4	Evidence $\times$ Generator	zero-provider qualified	automatic method
M5	untouched E3	sealed	confirmation

Table 2: Evidence ladder and current authority. Pending stages are not written as successful results.

Work	Main object	Update interface	Boundary
Rethinking Skills	feedback condition over rounds	LLM optimizer + validation	failure feedback and heterogeneity are prior findings
SkillRevise	repair weak authored skills	diagnosis + principles + LLM revision	structured diagnosis/repair are prior art
WikiSkill	accumulate experience across evolution	trace $\rightarrow$ wiki $\rightarrow$ skill	persistent intermediate knowledge is prior art
E2-R17	matched-evidence generator variance	free-form vs. canonical typed state	exact-evidence replay + Evidence $\times$ Generator test

Table 3: Closest novelty collisions and the residual object studied here.

6 RELATED WORK AND NOVELTY BOUNDARY

The closest feedback study is *Rethinking Self-Evolving Agent Skills*, which varies normal, failure-only, and success-only feedback across models and benchmarks and finds sparse, heterogeneous improvement (Liu et al., 2026b). It already establishes that failed trajectories can matter; our question begins where that result stops: under matched evidence, does the state-generation realization itself remain a material factor?

SkillRevise diagnoses execution-grounded defects, retrieves reusable repair principles, and applies anchored skill revisions (Liu et al., 2026a). Structured diagnosis and repair principles are therefore prior art, not our contribution. WikiSkill similarly separates raw traces, persistent accumulated knowledge, and executable skills (Tang et al., 2026); we do not claim experience-to-persistent-knowledge compilation as new. MindMemOS is the first-party skill-evolution substrate used by our free-form baseline (Liang et al., 2026). Other recent systems study behavioral validation, sustained evolution signals, and skill-library adaptation (He & Yang, 2026; Yan et al., 2026; Han et al., 2026).

Our residual novelty is narrower: *matched-evidence state-generation variance as a causal object*, followed by a prospective Evidence $\times$ Generator factorization. The typed compiler is not novel merely because it is structured; it matters only if controlling the state-writing interface prospectively changes utility or reproducibility relative to the native updater.

7 LIMITATIONS AND CONCLUSION

The strongest current localization is deliberately a selected development case. Two fresh exact-evidence updater realizations can falsify deterministic regeneration of the historical state, but they cannot estimate a population variance component. The typed vocabulary is also engineered for the current spreadsheet substrate and may align with its evaluator. Our score-only generic falsifier and fresh suite address this alternative only within the controlled domain. Finally, the compiler differs from free-form synthesis in determinism, canonicalization, scope, redundancy, and state length, so a future positive bridge result is a complete state-generation-method effect rather than an isolated determinism effect. Hosted temperature-zero calls may also retain residual nondeterminism; outcome-conditioned redraws are therefore forbidden rather than averaged away.

The scientific lesson is nevertheless concrete: experience reaches future behavior only through a persistent artifact. In our development chain, a global rejected-witness intervention is heterogeneous, a useful state survives when frozen, and byte-identical evidence does not reliably regenerate its historical effect through fresh free-form synthesis. The evidence and the state are therefore distinct experimental objects. Separating evidence selection, state generation, state semantics, and downstream behavior provides a sharper diagnosis of self-evolution failures and a falsifiable route toward variance-controlled persistent-state generation.

REPRODUCIBILITY STATEMENT

All scientific objects are content-addressed and bound to frozen task, model, updater, verifier, provider-budget, and execution-order contracts. Completion audits precede outcome analysis, interrupted runs are resumed without replaying completed units, and prospective SCREEN/VALIDATION panels are disjoint. Appendix A records the key runtime, suite, recovery, and compiler invariants.

AI USE STATEMENT

Generative AI tools were used to assist with hypothesis refinement, methodology critique, experiment design, implementation, literature discovery, and manuscript drafting/editing. AI-assisted code and scientific artifacts were executed or checked under the same explicit test, provenance, and review gates described in the paper; independent model-based reviews were treated as design feedback rather than scientific outcomes. The authors reviewed the resulting claims, code, analyses, and manuscript and take responsibility for the final content.

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A SCIENTIFIC LINEAGE AND EXECUTION INTEGRITY

Closed primary study. The closed DeepSeek Repair2 object contains 48 pairs, 96 learned states, and 1728 heldout units. Completion was audited before score analysis. The final scientific verdict is immutable and remains `HOLD_MRW_UNDERPOWERED_OR_HETEROGENEOUS`.

Development-only mechanism studies. The S1 witness-selector pilot, frozen-state stability re-measurement, exact-evidence replay, and G0–G3 deterministic state intervention are explicitly development objects. They are not pooled with the closed 48-pair effect to manufacture a confirmatory estimate. The exact-evidence replay verdict is recorded as `FIRST_FAIL_EXACT_EVIDENCE_UPDATER_REPLICATION_FAIL_STATE_GENERATION_VARIANCE` in the machine-readable lineage; the long identifier is not used as prose in the main paper.

Exactly-once recovery. The G0–G3 micro was interrupted by a hosted weekly quota boundary after 45/72 valid measurements. Recovery V3 inherits those 45 completed units and contains exactly 27 unresolved units in the original hash-balanced task/arm order. Completed units are never replayed. One task is split across provider eras; a pre-outcome 17-task sensitivity excluding that task may veto but never rescue the original 18-task gate.

No partial-result adaptation. At manuscript revision time, the 27 remaining G0–G3 units are pending and no partial state-level effect has been read. The automatic compiler vocabulary and bridge design were developed as zero-provider proposal objects. They cannot be modified using the pending G0–G3 outcomes and then treated as prospectively frozen under the same scientific object.

B FROZEN DETERMINISTIC STATE SEMANTICS

The state-level intervention uses the exact initial spreadsheet skill plus the following incremental procedural semantics.

G1: output verification. After saving `output.xlsx`, reload it with `load_workbook("output.xlsx", data_only=True)` and verify the exact target cells before stopping. If verification fails, fix and save again.

G2: completion loop. For each spreadsheet task: (1) inspect workbook structure and target cells, (2) read required source data, (3) compute requested values, (4) materialize values into exact target cells, (5) save `output.xlsx` without modifying the input, and (6) reload and verify the target cells. Inspection alone is not completion.

G3: tool-error recovery. In addition to G2, if a shell, Python, or tool invocation fails or is malformed, issue a clean corrected command rather than stopping or repeating the same broken call unchanged, then continue through write, save, reload, and verification.

The automatic compiler’s canonical primitive surfaces reproduce these frozen state texts byte-for-byte for corresponding typed diagnoses.

C BRIDGE SUITE QUALIFICATION

The fresh bridge suite is generated only in b7–b9 and is disjoint from the prior controlled suite by both task ID and XLSX SHA-256. Candidate count is 162; 120 tasks are formal scientific tasks and 42 are integrity-only reserves. The formal set contains 96 update tasks, 12 SCREEN heldout tasks, and 12 VALIDATION heldout tasks.

The zero-provider qualification records:

- logical dataset SHA-256: 25785a89d25424d7e8ed2d75d802f7e968a785d318b521272c38929b36342f05;
- SCREEN/VALIDATION heldout overlap: 0;
- prior/new task-ID overlap: 0;
- prior/new XLSX SHA-256 overlap: 0;
- deterministic rebuild match: PASS;
- generation runtime: CPython 3.12.3, openpyxl 3.1.5, zlib 1.3.

D PROSPECTIVE INTERPRETATION MATRIX

Generator contrast	Evidence contrast	Allowed interpretation
Fail	any	State-generation bottleneck method story stops
Pass	Fail	Keep reliable/canonical state compilation; drop unique rejected-failure claim
Pass	Pass, but generic falsifier fails	Generic/score-conditioned workflow compilation; no trajectory-conditioned diagnosis claim
Pass	Pass, generic falsifier passes, interaction passes	Eligible for separate untouched-E3 confirmation proposal

No cell in this matrix automatically authorizes E3, a second backbone, a public benchmark, or paper-level generalization.